

Rndom Forest

```
[78]: from sklearn.ensemble import RandomForestClassifier
```

```
rf_model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", RandomForestClassifier(
        n_estimators=300,
        random_state=42,
        n_jobs=-1
    ))
])
```

```
[79]: rf_model.fit(X_train, y_train)
```

```
y_pred_rf = rf_model.predict(X_test)
y_proba_rf = rf_model.predict_proba(X_test)[: , 1]
```

```
[80]: # Evaluate
```

```
[81]: print("Accuracy:", accuracy_score(y_test, y_pred_rf))
```

```
print("\nClassification Report:")
print(classification_report(y_test, y_pred_rf))
```

```
print("ROC-AUC:", roc_auc_score(y_test, y_proba_rf))
```

Accuracy: 0.7821149751596878

Classification Report:

	precision	recall	f1-score	support
No	0.83	0.89	0.86	1035
Yes	0.61	0.48	0.54	374
accuracy			0.78	1409
macro avg	0.72	0.69	0.70	1409
weighted avg	0.77	0.78	0.77	1409

ROC-AUC: 0.8194864243457594